DO NOW — RECALL

Lesson 105 · RAM, ROM and Virtual Storage: Different R

Name: _____ Class: _____ Date: _____

1. Where are running program instructions and data commonly held?

2. What does ROM or firmware flash retain?

3. What is a remote network drive an example of?

4. Rising edge D-type flip-flop D0 Q1: Q just after edge?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

RAM, plus CPU caches/registers.

2. Model answer

Startup/firmware instructions without power.

3. Model answer

Virtual storage.

4. Model answer

0, because D is sampled at the active edge.

By the end, you can…

• RAM holds currently used programs/data and normally loses content without power; ROM or firmware flash retains startup routines.

• OCR virtual storage means logical storage presented from physically remote or pooled devices, such as a network drive or cloud service.

• Do not confuse virtual storage with virtual memory. Virtual memory uses secondary storage to extend the apparent addressable memory of processes; virtual storage is remote-file presentation.

Vocabulary — know these terms

RAM — Volatile working memory.

ROM — Non-volatile storage for relatively fixed code.

virtual storage — Remote/pooled physical storage presented logically.

virtual memory — OS memory abstraction using paging.

NEW LEARNING

Read, annotate and explain

Understand

RAM holds currently used programs/data and normally loses content without power; ROM or firmware flash retains startup routines.

Apply

OCR virtual storage means logical storage presented from physically remote or pooled devices, such as a network drive or cloud service.

Evaluate

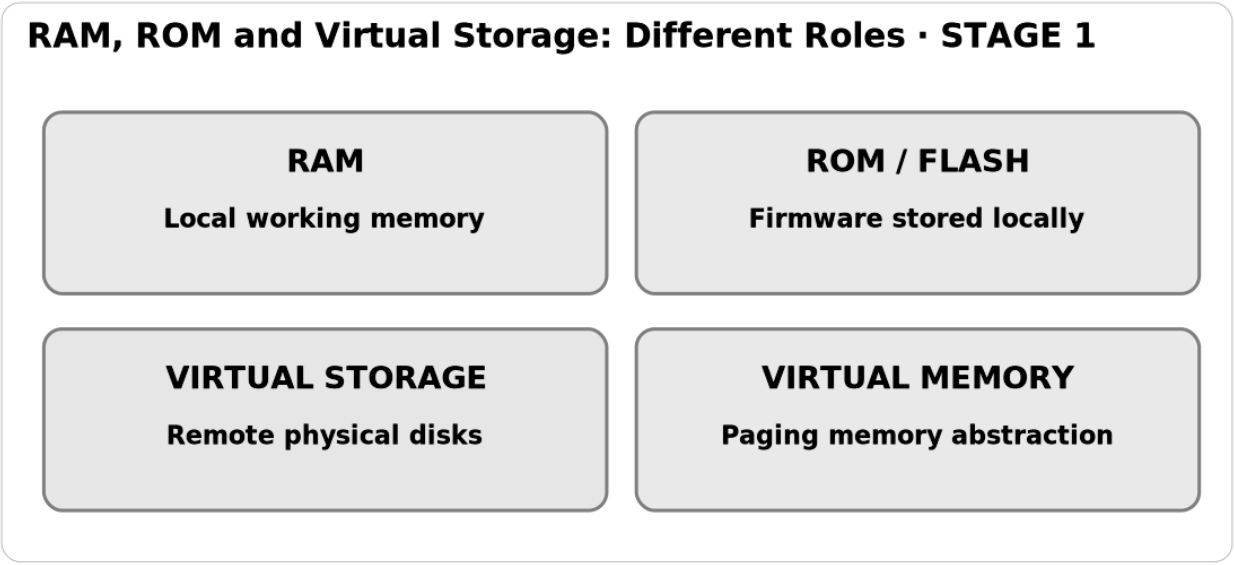
Do not confuse virtual storage with virtual memory. Virtual memory uses secondary storage to extend the apparent addressable memory of processes; virtual storage is remote-file presentation.

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

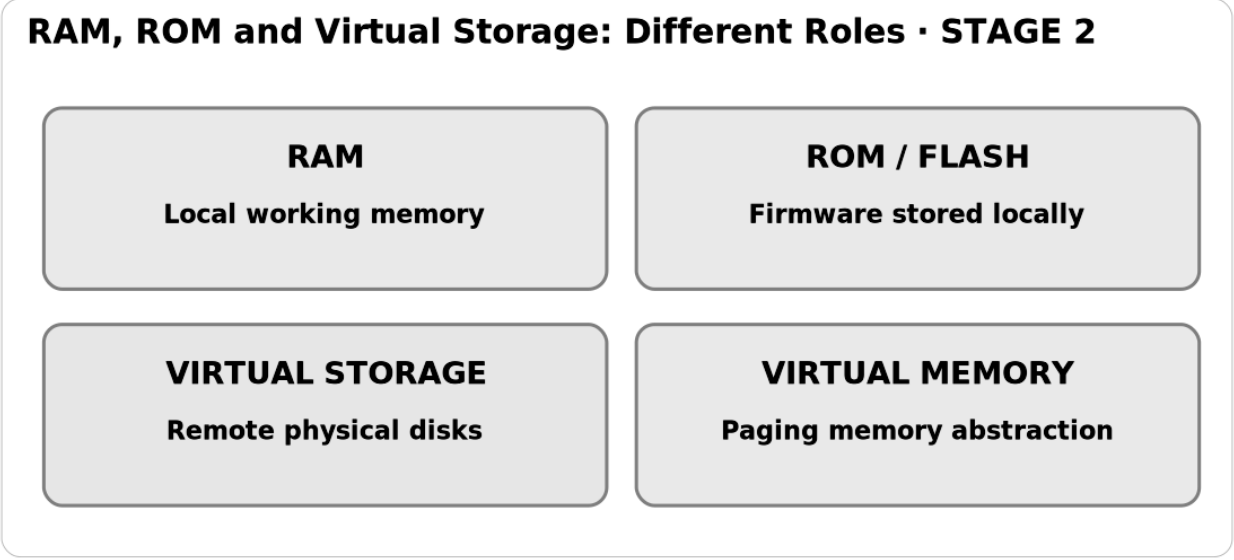
Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



RAM, ROM and Virtual Storage: Different Roles · STAGE 1; RAM: Working memory; volatile

New Learning

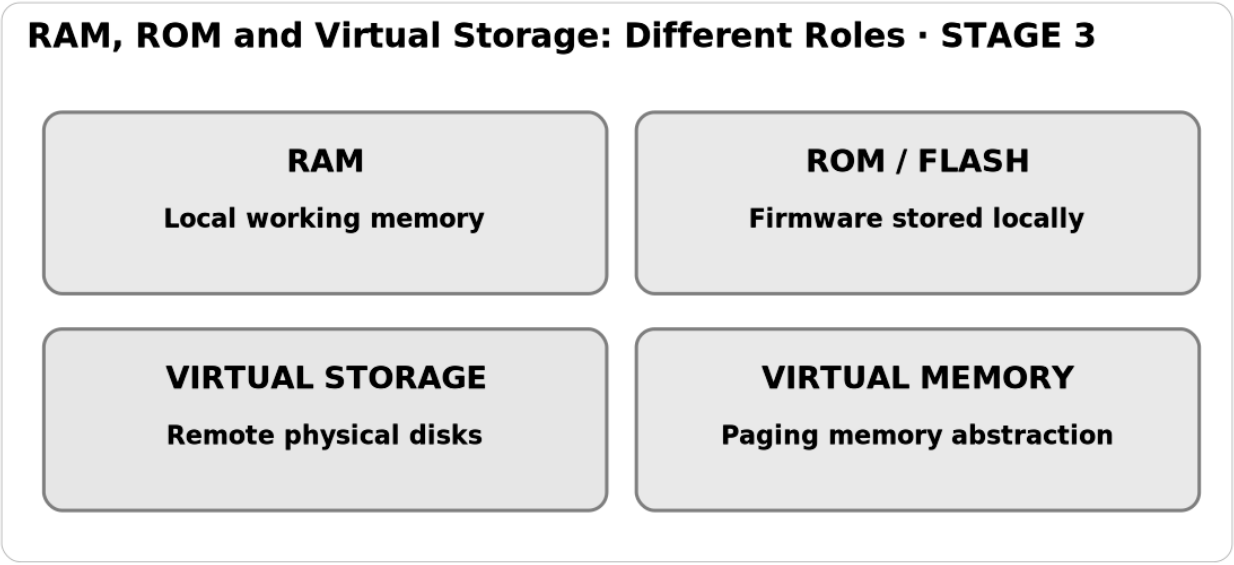


RAM, ROM and Virtual Storage: Different Roles · STAGE 2; RAM: Working memory; volatile; ROM / FIRMWARE FLASH: Retains startup firmware without power

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

New Learning

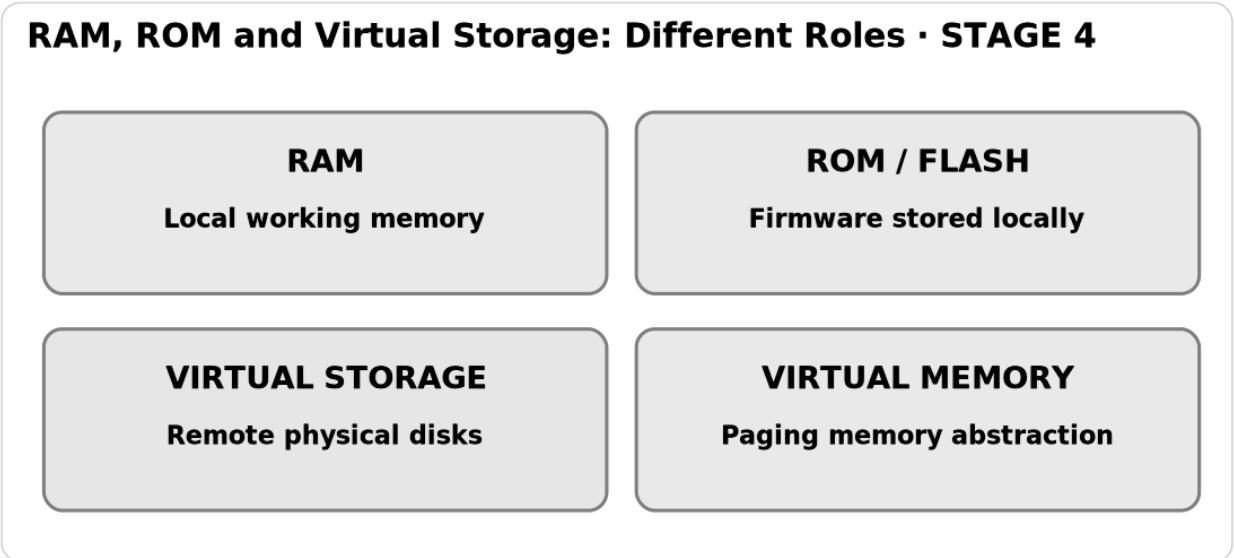


RAM, ROM and Virtual Storage: Different Roles · STAGE 3; RAM: Working memory; volatile; ROM / FIRMWARE FLASH: Retains startup firmware without power; VIRTUAL STORAGE: Remote physical disks appear as accessible drive

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



RAM, ROM and Virtual Storage: Different Roles · STAGE 4; RAM: Working memory; volatile; ROM / FIRMWARE FLASH: Retains startup firmware without power; VIRTUAL STORAGE: Remote physical disks appear as accessible drive; VIRTUAL MEMORY: Separate OS concept: paging to secondary storage

I DO — WORKED EXAMPLE

Follow the worked method

A pupil opens a document from a mounted school network drive. Where is the file physically stored?

Explain the choice

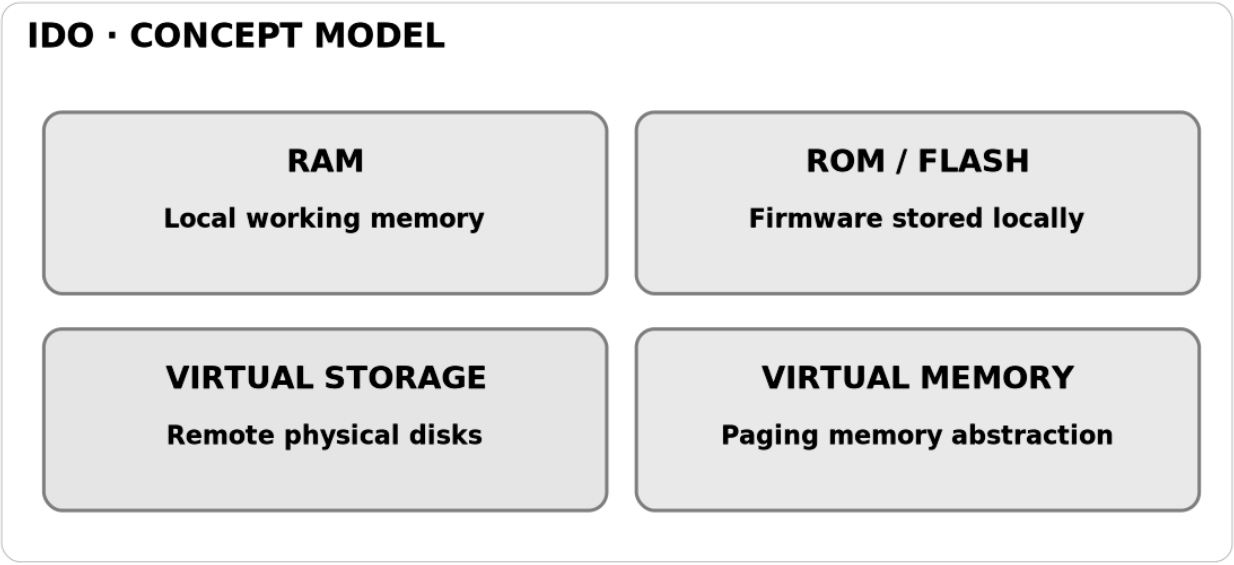
On one or more remote network storage devices, even if Finder/Explorer shows a drive like local storage.

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

A pupil opens a document from a mounted school network drive. Where is the file physically stored?

On one or more remote network storage devices, even if Finder/Explorer shows a drive like local storage.



IDO · CONCEPT MODEL; A pupil opens a document from a mounted school network drive. Where is the file physically stored?: On one or more remote network storage devices, even if Finder/Explorer shows a drive like local storage.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

TRY: Attempt the guided questions in BLACK PEN. DISCUSS: Compare approaches with a partner, explain your reasoning and agree one improvement. IMPROVE: After the Exit Ticket, use the source CHECK model in GREEN PEN to correct and explain errors.

Guided 1 — TRY

A laptop runs out of RAM and the OS moves inactive pages to an SSD. Is this OCR 1.1.3(d) virtual storage?

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations or a justified explanation.

Question 1

Explain the role and volatility of RAM and ROM/firmware storage. [6]

Question 2

Define virtual storage and give an example, two benefits and a disadvantage. [8]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations or a justified explanation.

Question 3

Correct a pupil's claim that 'virtual storage is extra RAM created when memory fills up'. [6]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking.

OCR H446 1.1.3(d) virtual storage is another term for swapping RAM pages.

Circle: Confident / Getting there / Need help

Explain what you can now do independently.

GREEN PEN — Correct and explain

Choose one error. Write the correct method and explain precisely why your first response was wrong.

WE DO — CHECK AND IMPROVE

Source-grounded model answers

CHECK only after completing the guided practice and Exit Ticket. Use GREEN PEN to improve your method and explain any error.

Guided 1 — CHECK

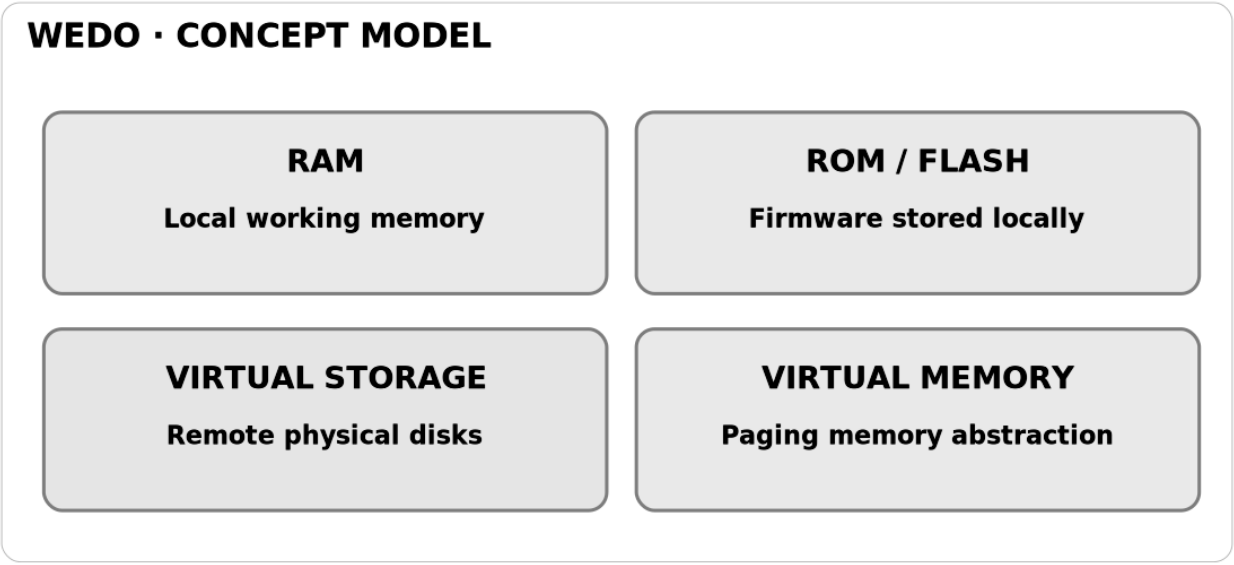
Explain the choice
No. It is virtual memory under 1.2.1, using secondary storage for memory pages.

We Do source CHECK · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

A laptop runs out of RAM and the OS moves inactive pages to an SSD. Is this OCR 1.1.3(d) virtual storage?

No. It is virtual memory under 1.2.1, using secondary storage for memory pages.



WEDO · CONCEPT MODEL; A laptop runs out of RAM and the OS moves inactive pages to an SSD. Is this OCR 1.1.3(d) virtual storage?: No. It is virtual memory under 1.2.1, using secondary storage for memory pages.